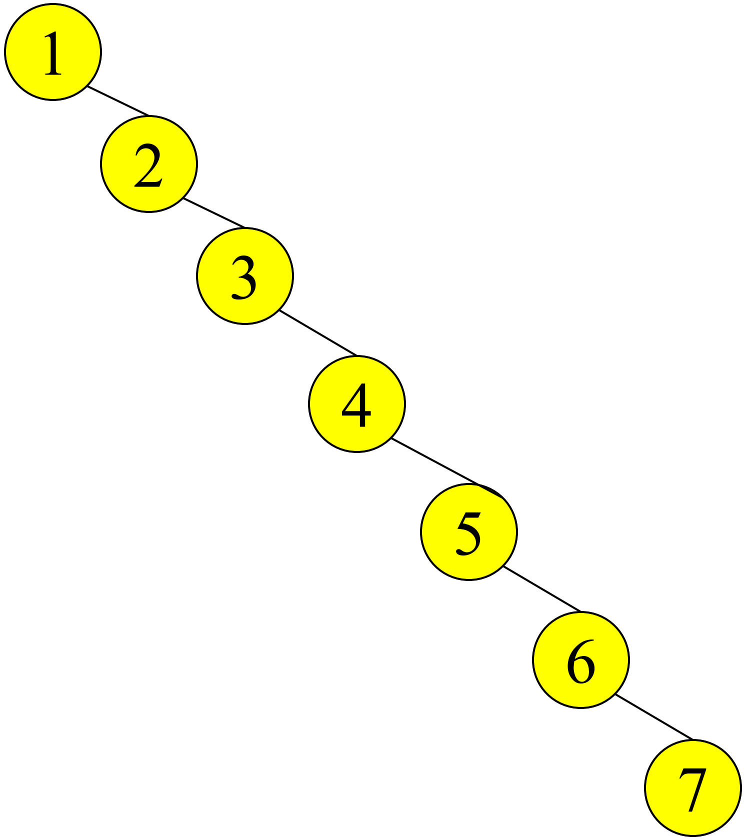
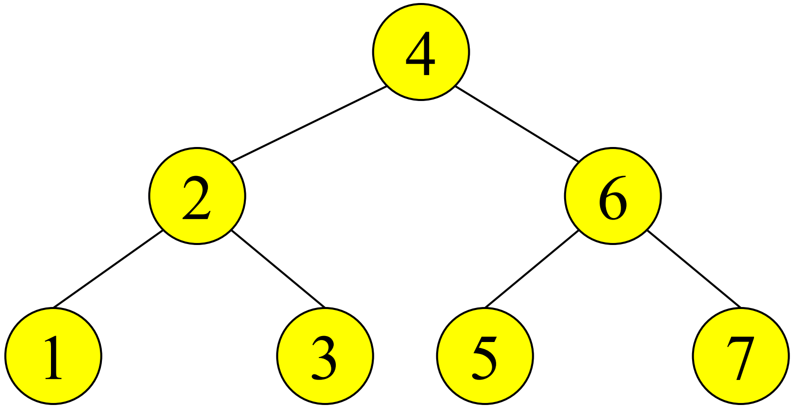
Arbori binari de cautare

Constructie arbore binar de cautare: 1, 2, 3, 4, 5, 6, 7

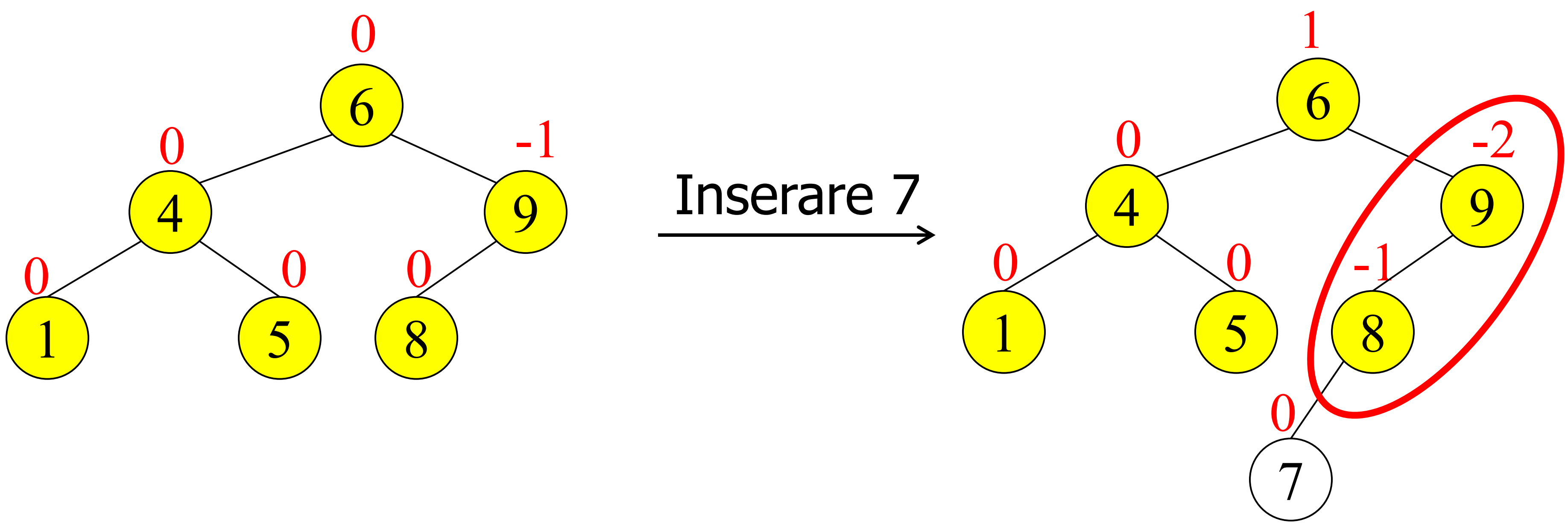




Arbore echilibrat

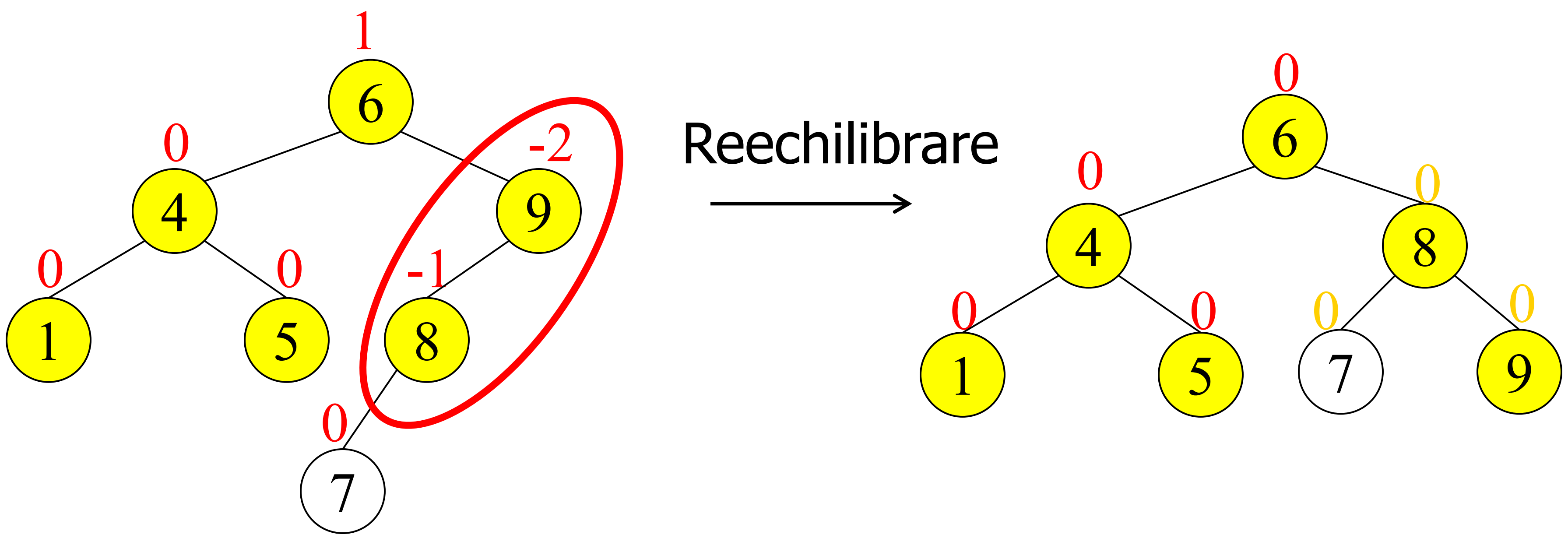
Arbori binari de cautare AVL

- Pentru orice nod diferenta intre inaltimile celor 2 subarbori este < 2
- Inserarea si eliminarea pot necesita reechilibrarea arborelui; in acest scop se utilizeaza factorul de echilibru = H(dr) – H(st)



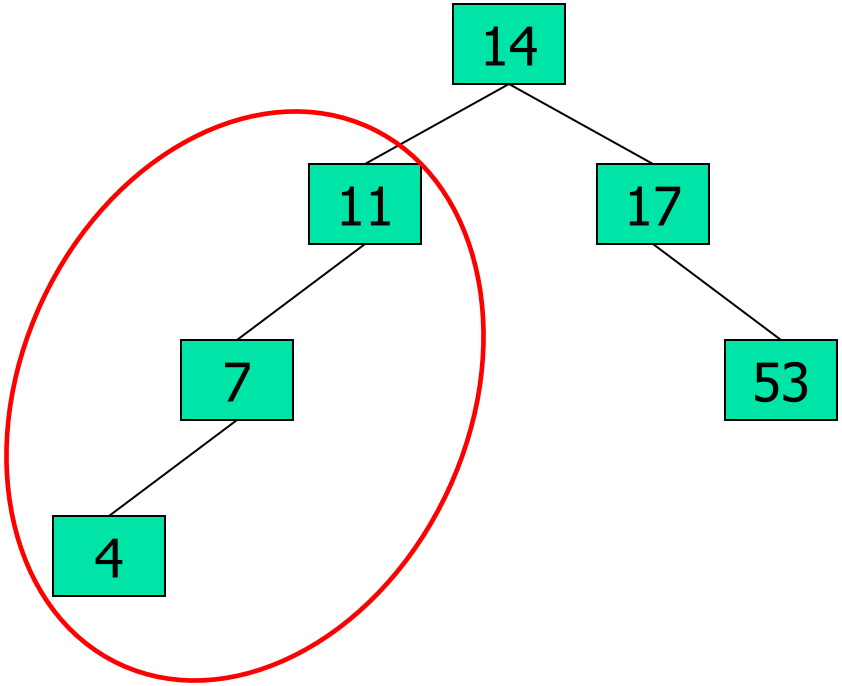
Arbori binari de cautare echilibrati - AVL

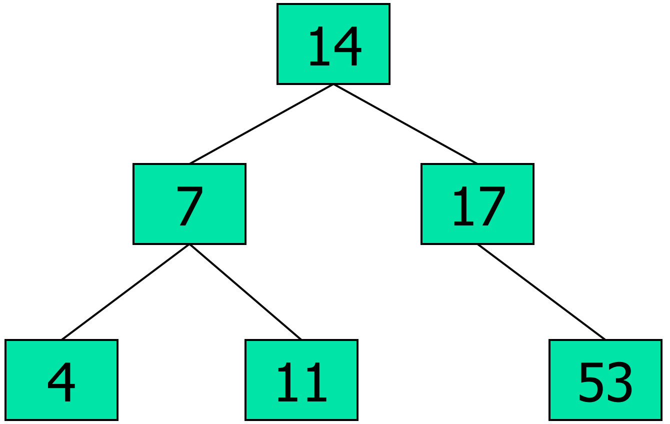
- Pentru orice nod diferenta intre inaltimile celor 2 subarbori este < 2
- Inserarea si eliminarea pot necesita reechilibrarea arborelui; in acest scop se utilizeaza factorul de echilibru = H(dr) – H(st)



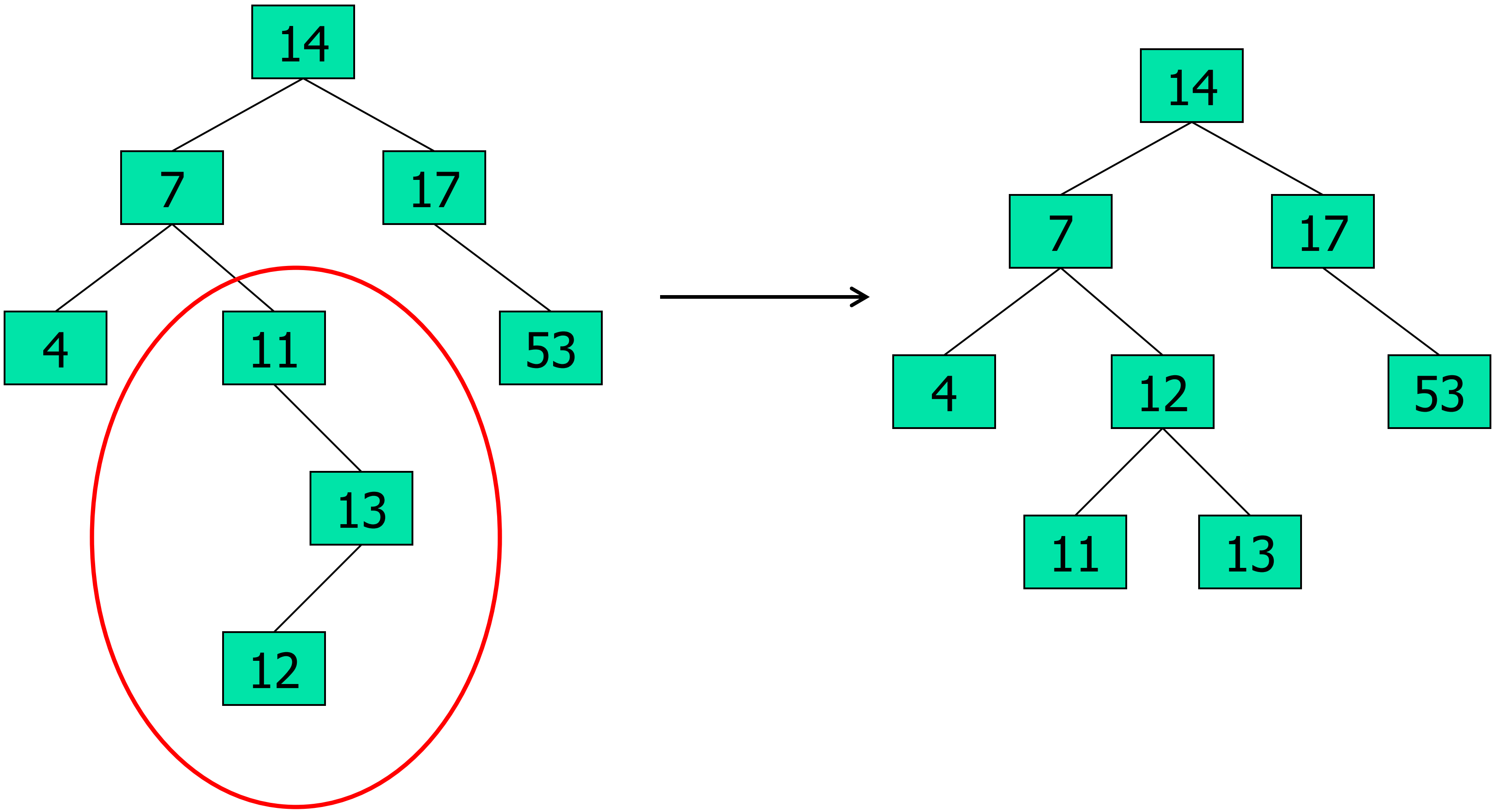
Arbori binari de cautare echilibrati - AVL

Inserare: **14, 17, 11, 7, 53, 4**

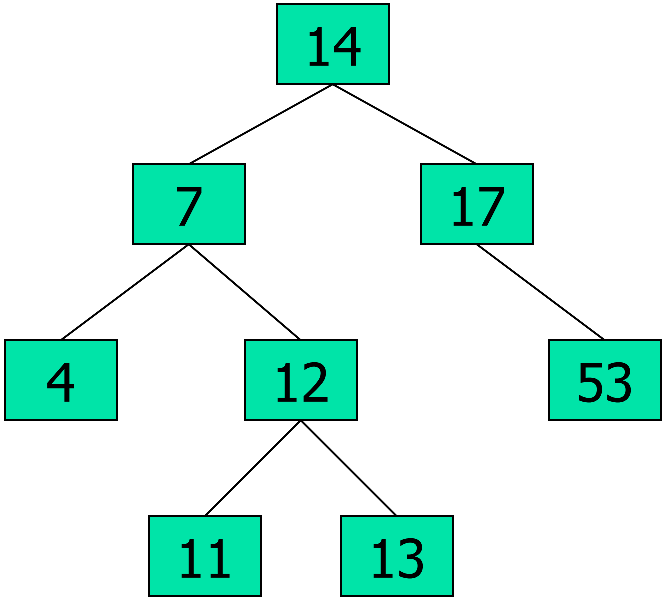




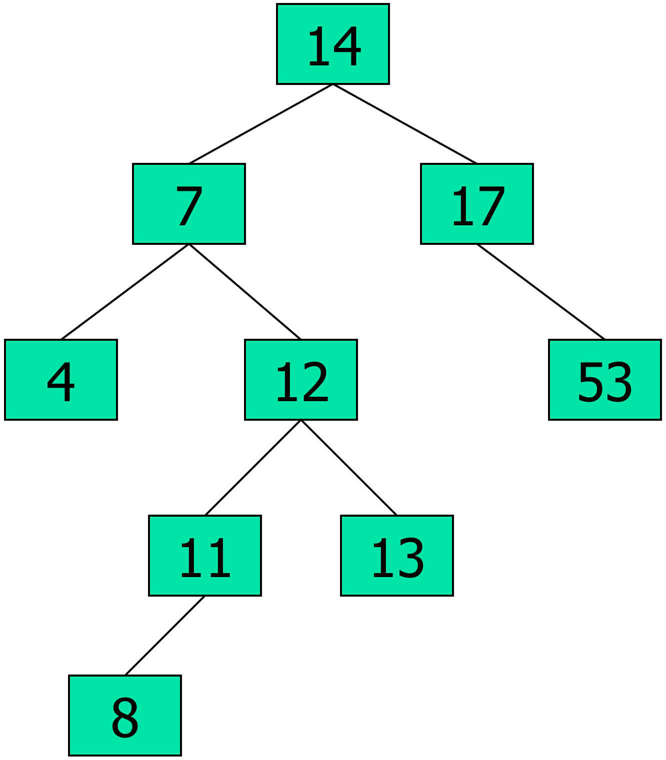
Arbori binari de cautare echilibrati - AVL



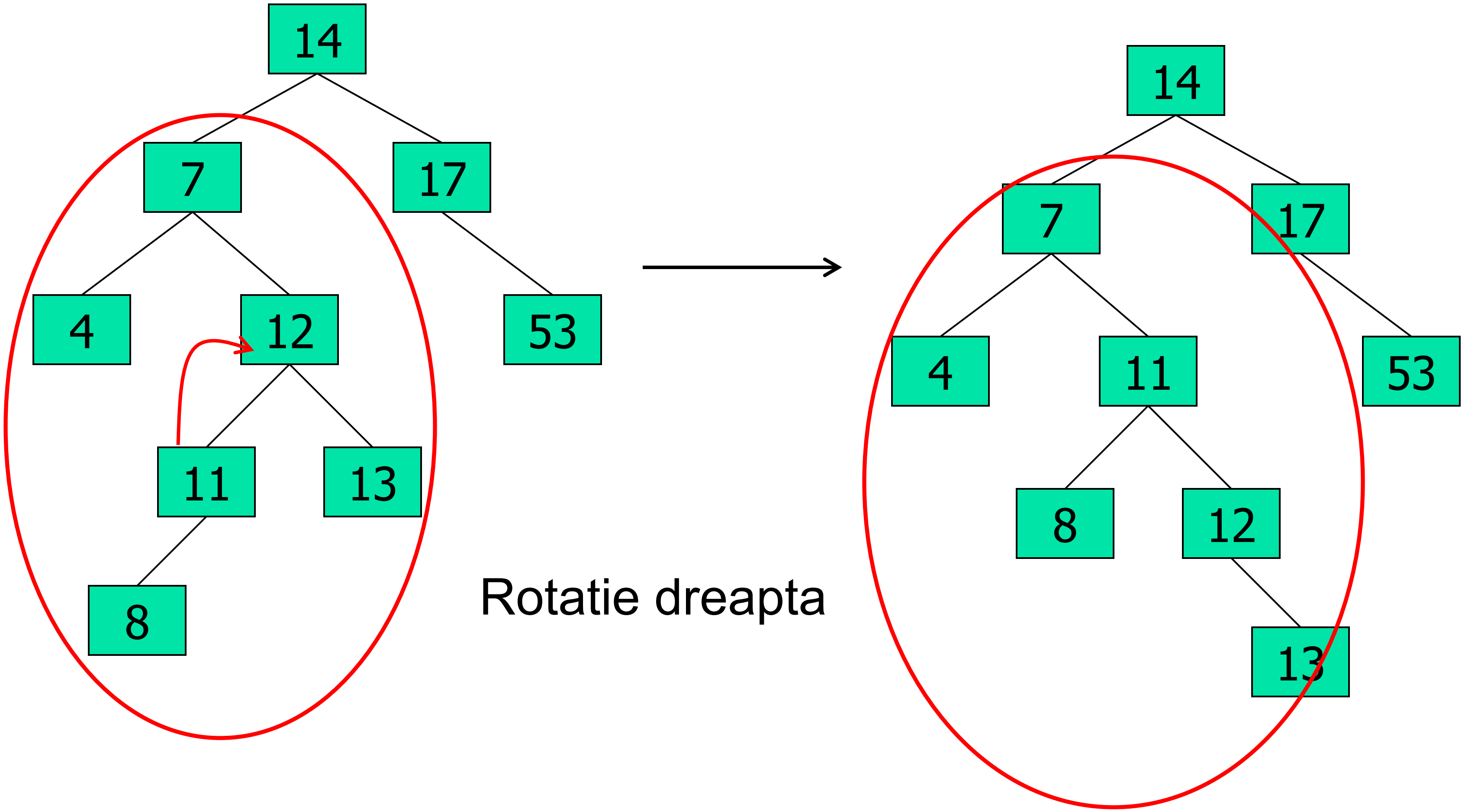
Arbori binari de cautare echilibrati - AVL



Inserare 8

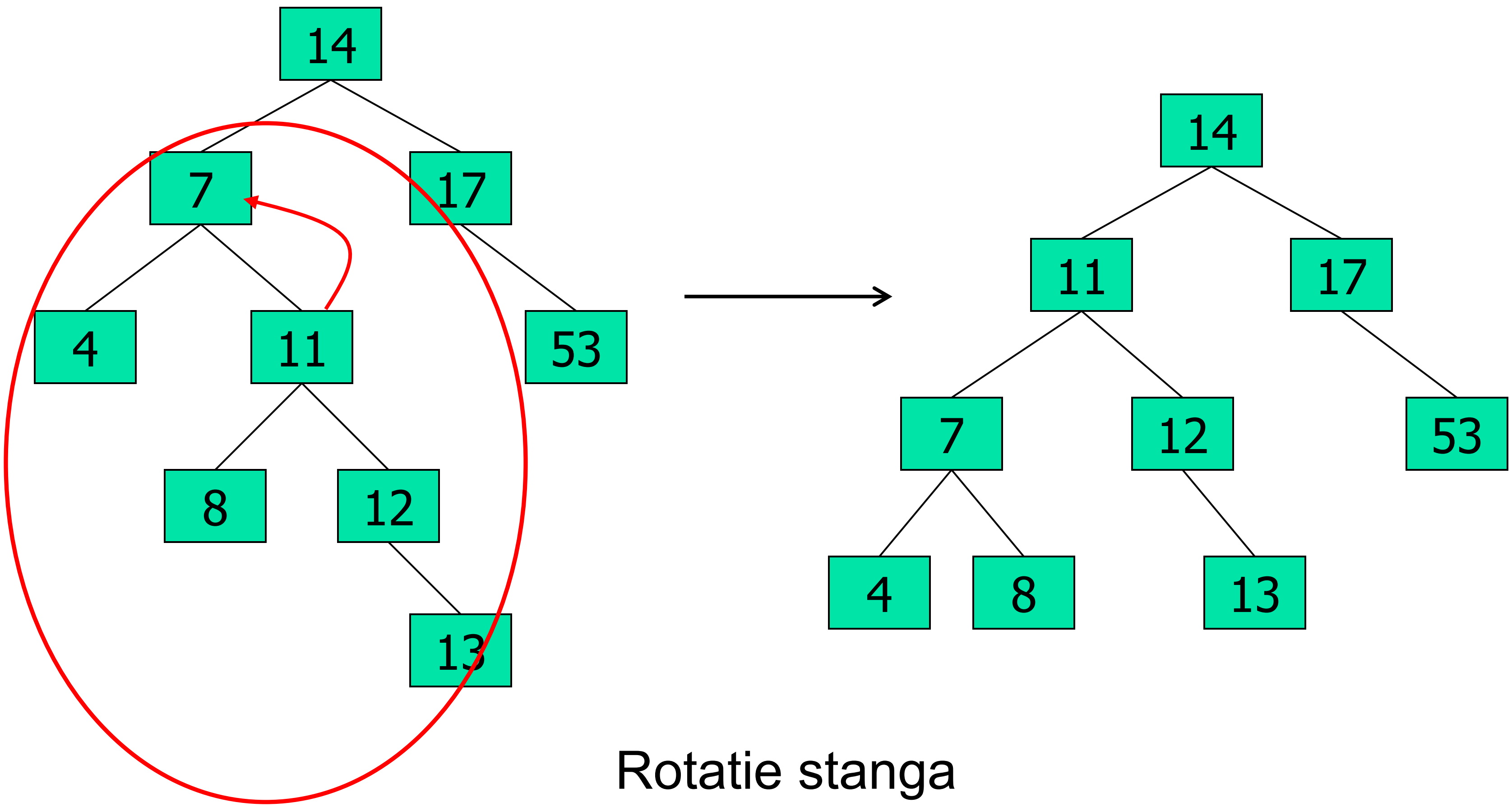


Arbori binari de cautare echilibrati - AVL



Rotatie dreapta

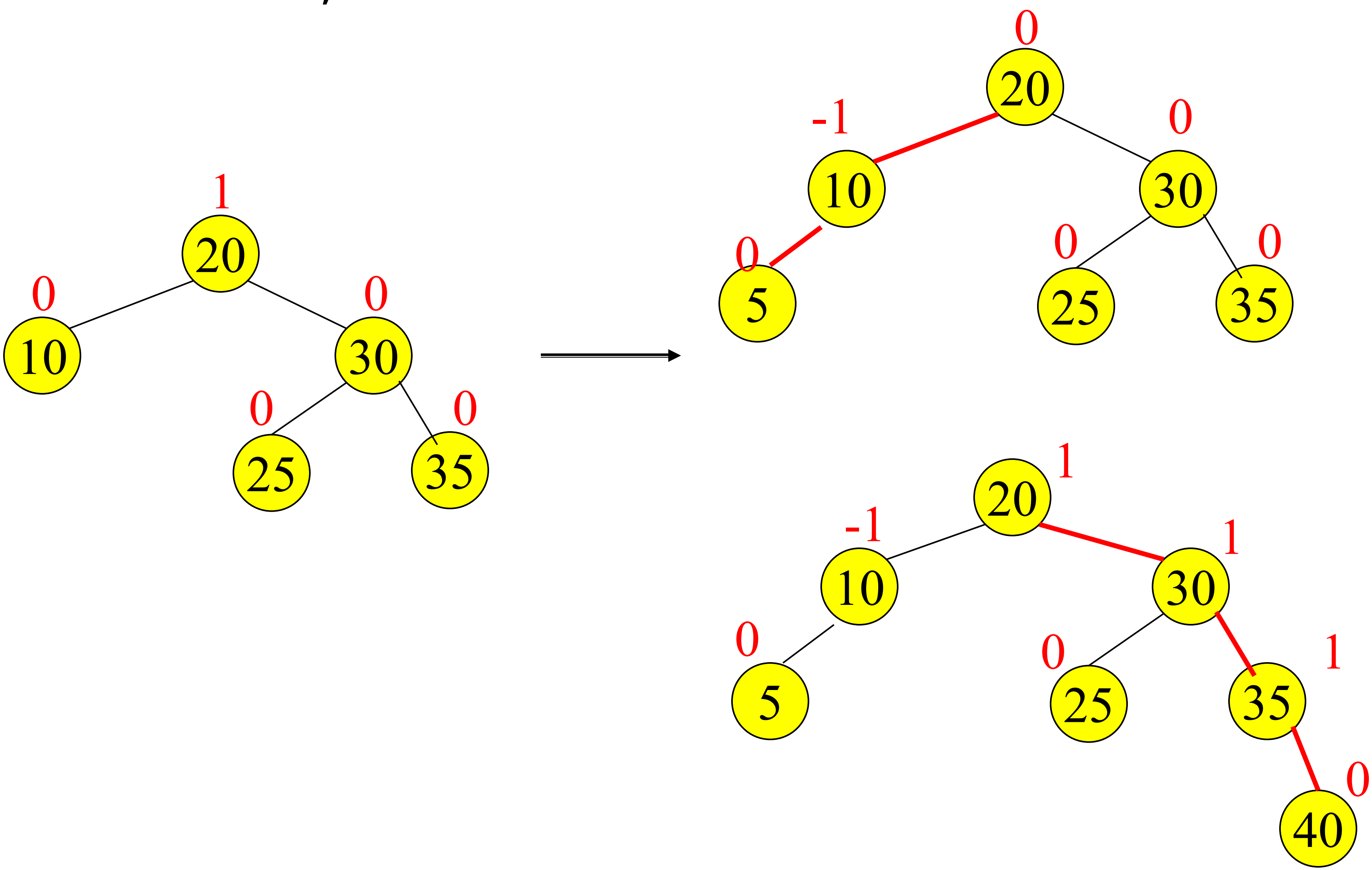
Arbori binari de cautare echilibrati - AVL



Rotatie stanga

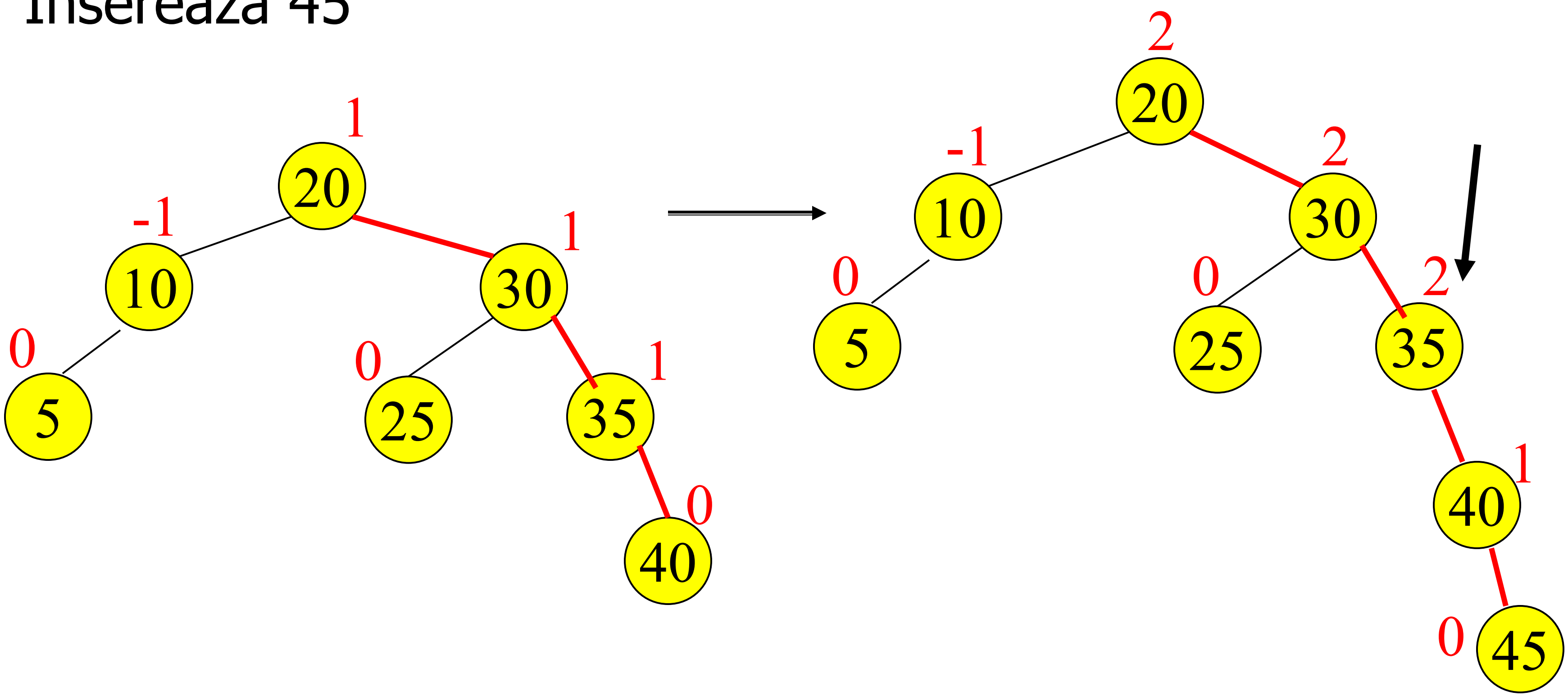
Exemplu: Arbori AVL

Insereaza 5, 40



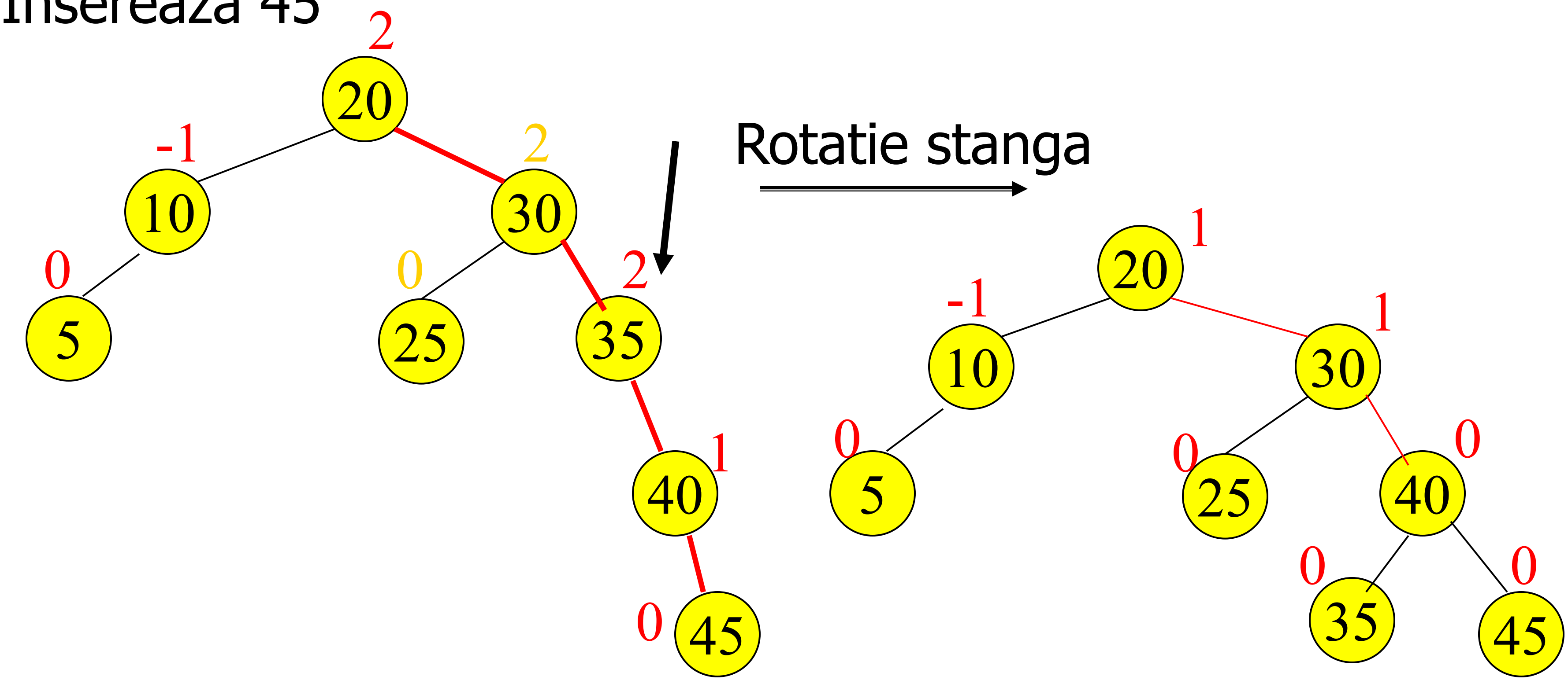
Exemplu: Arbori AVL

Insereaza 45



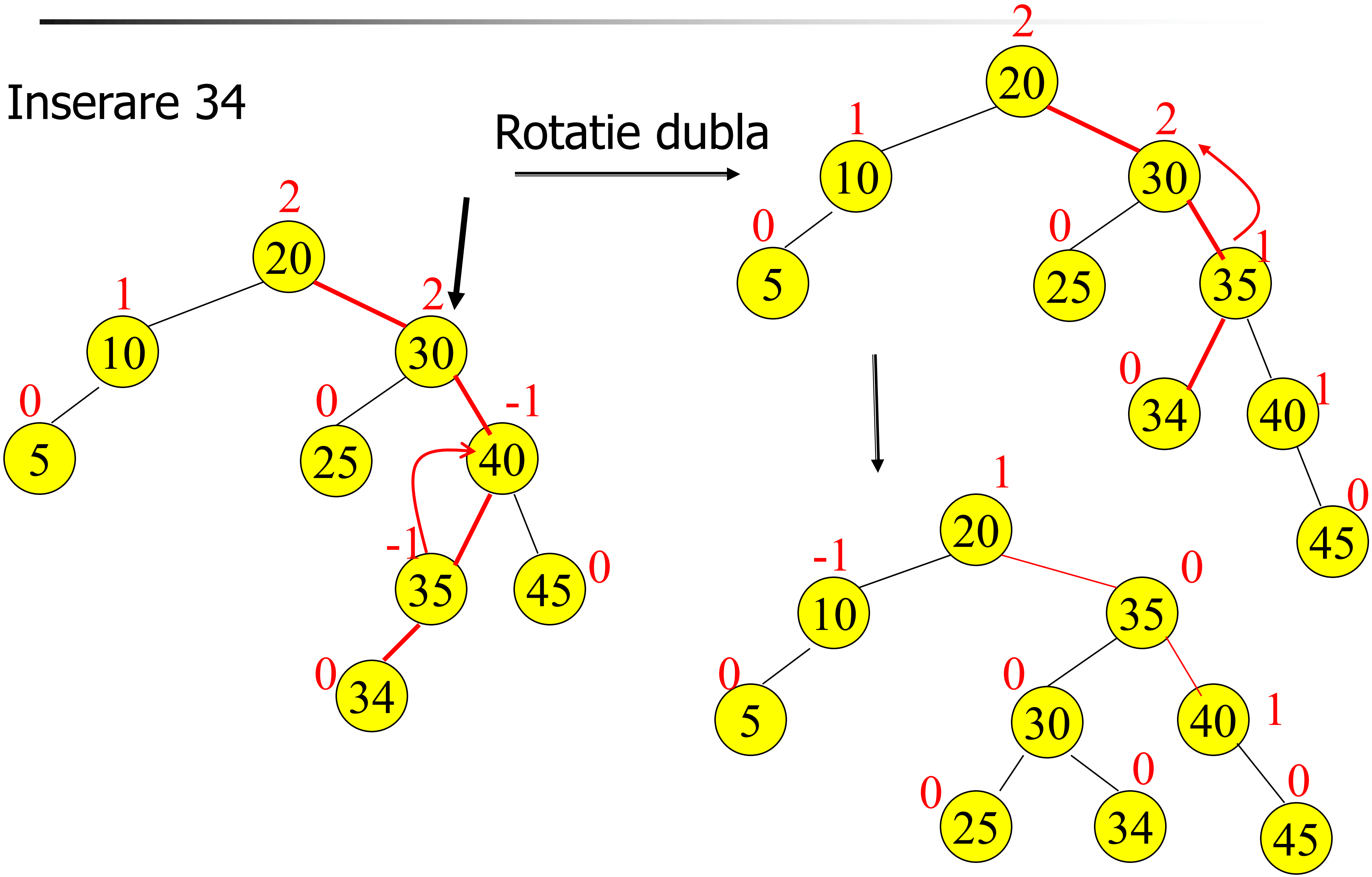
Exemplu: Arbori AVL

Insereaza 45

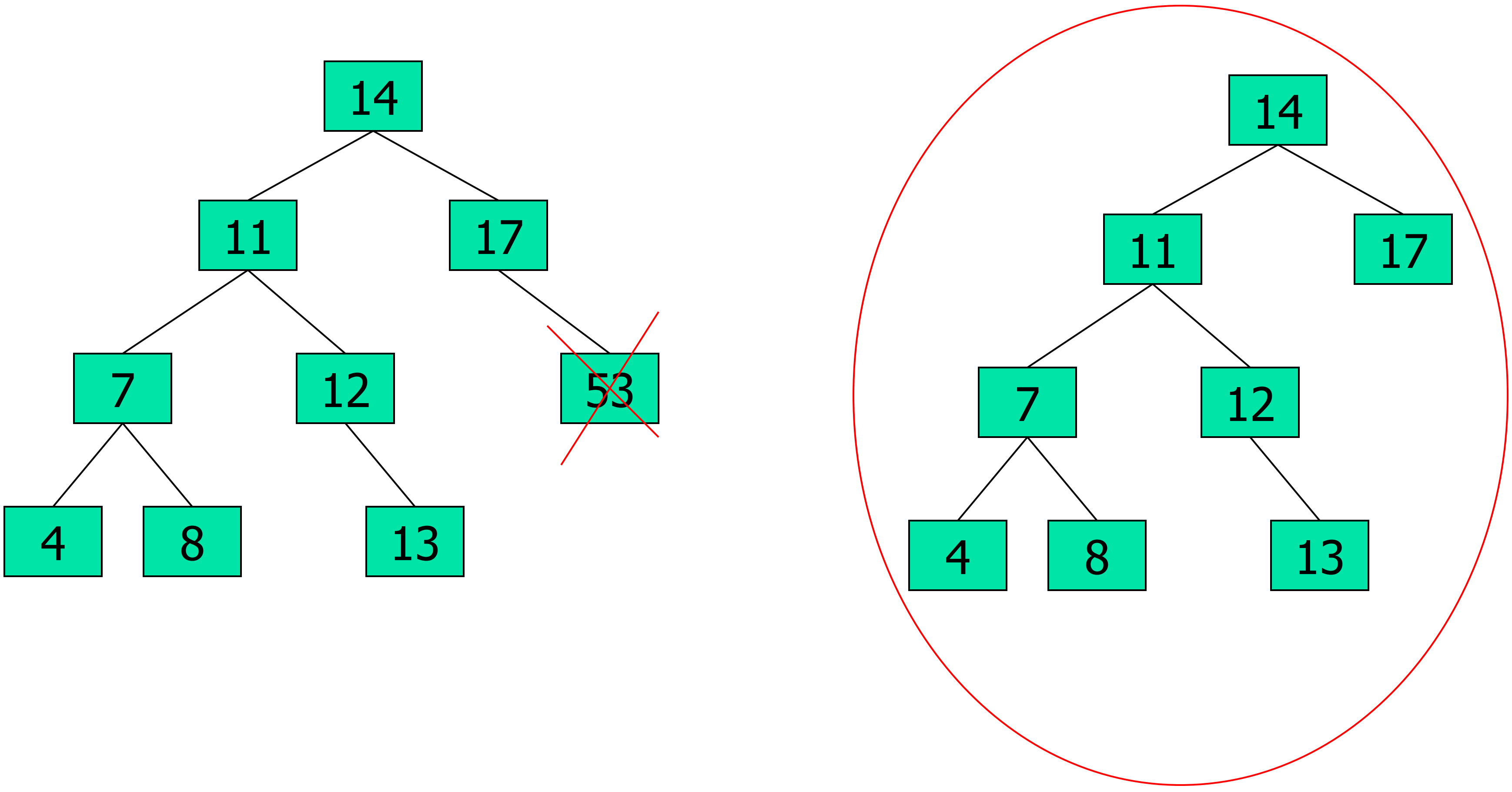


Exemplu: Arbori AVL

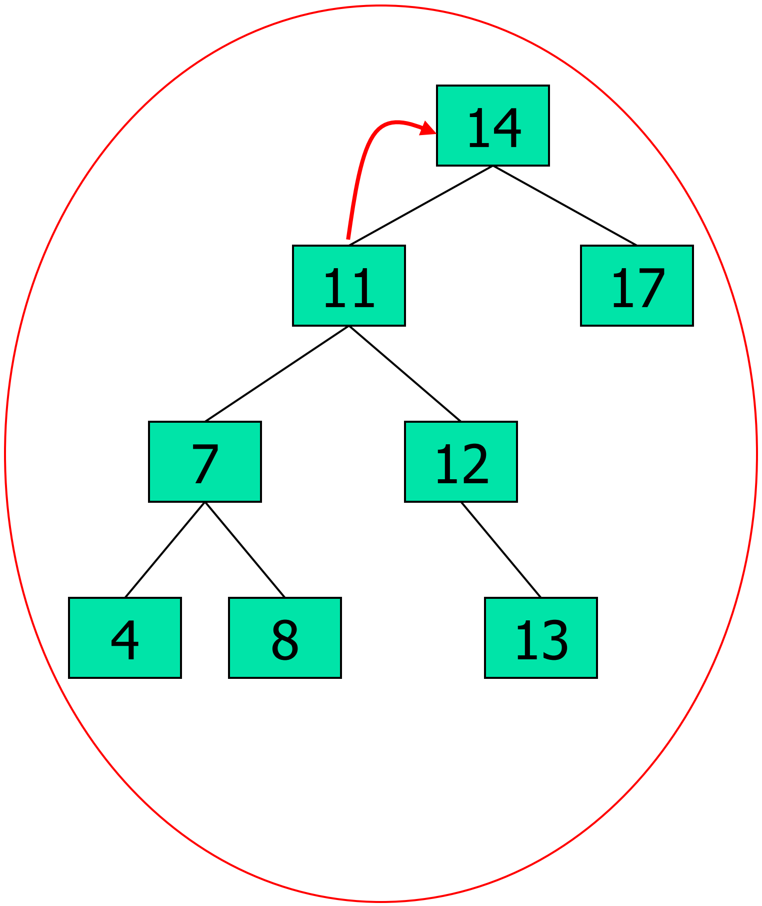
Inserare 34

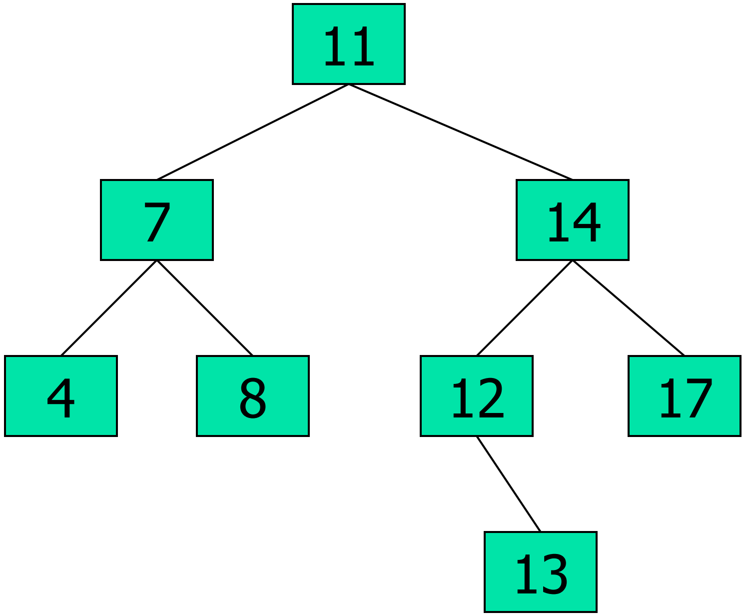


Eliminare din arbori AVL

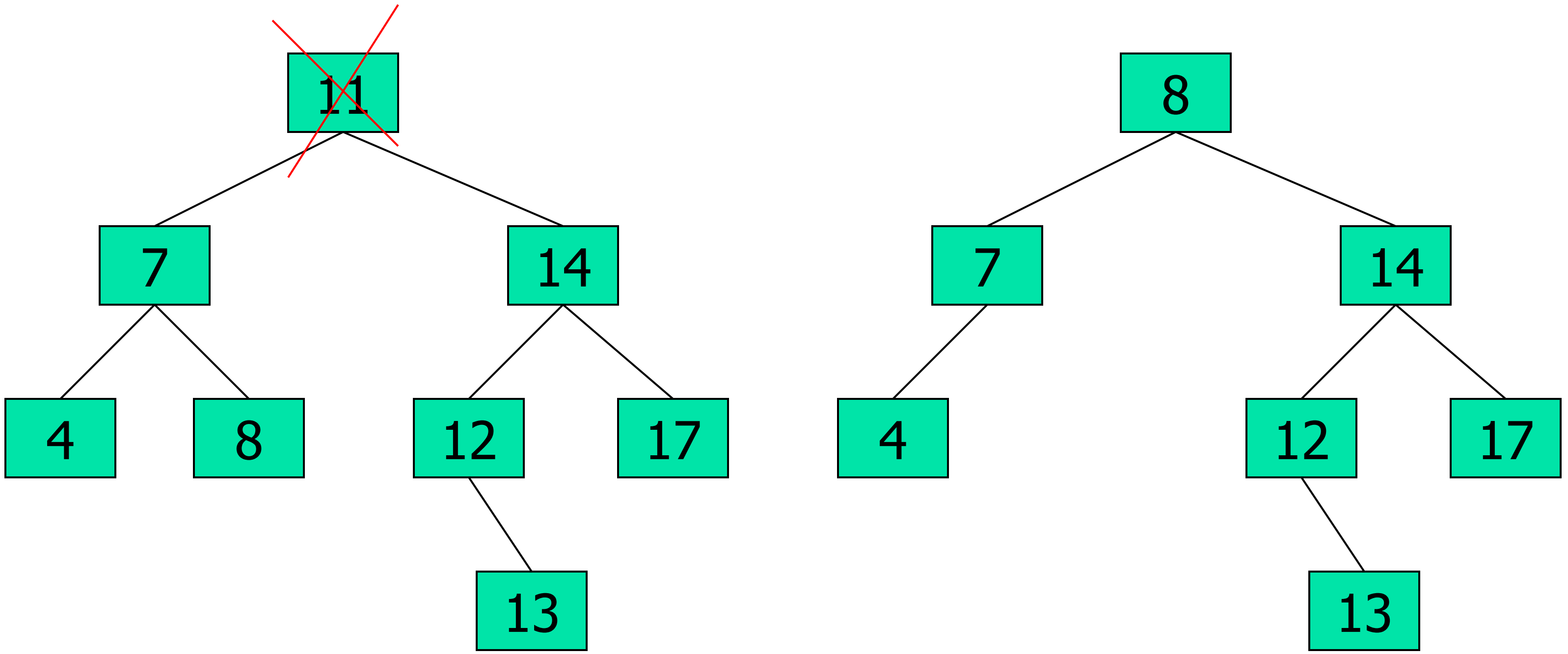


Eliminare din arbori AVL

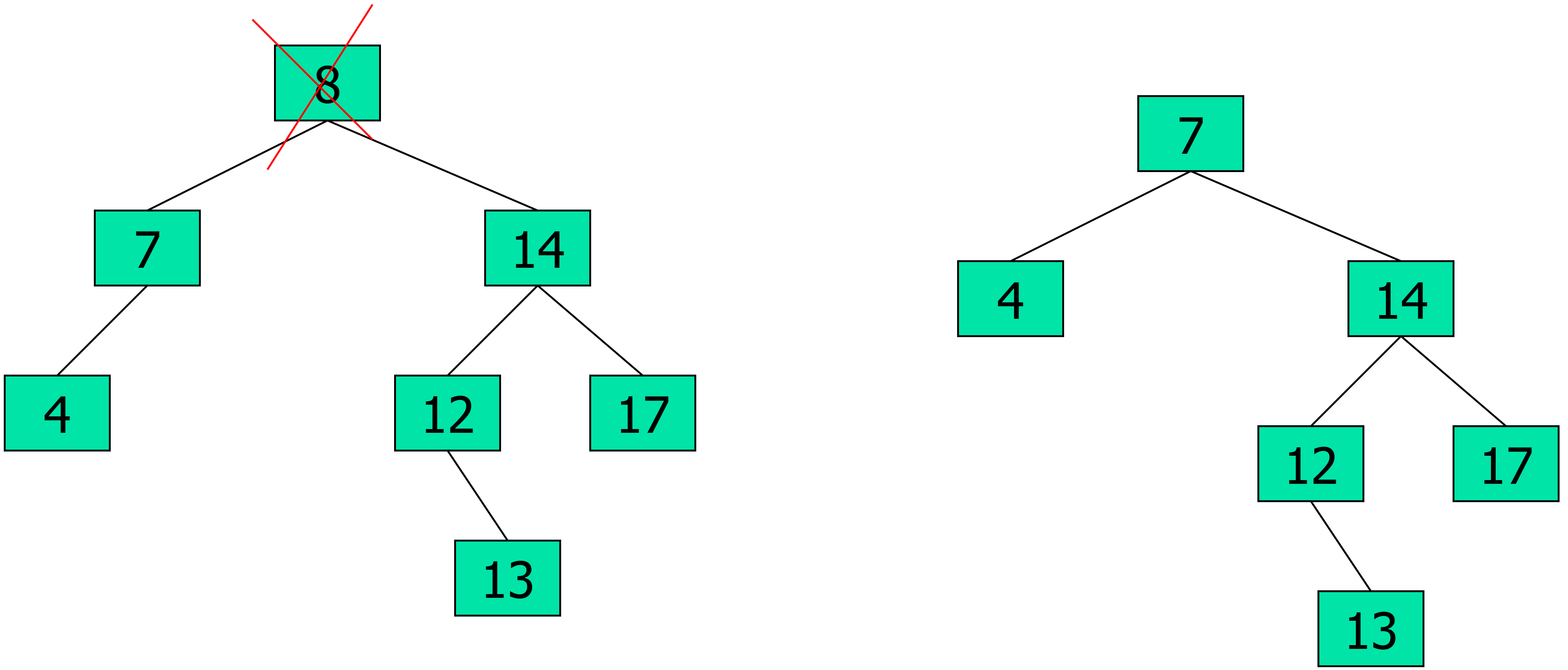




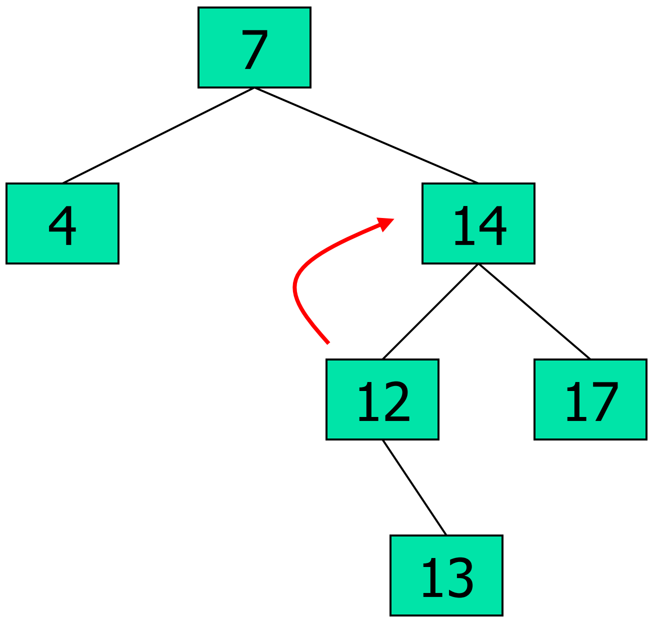
Eliminare din arbori AVL

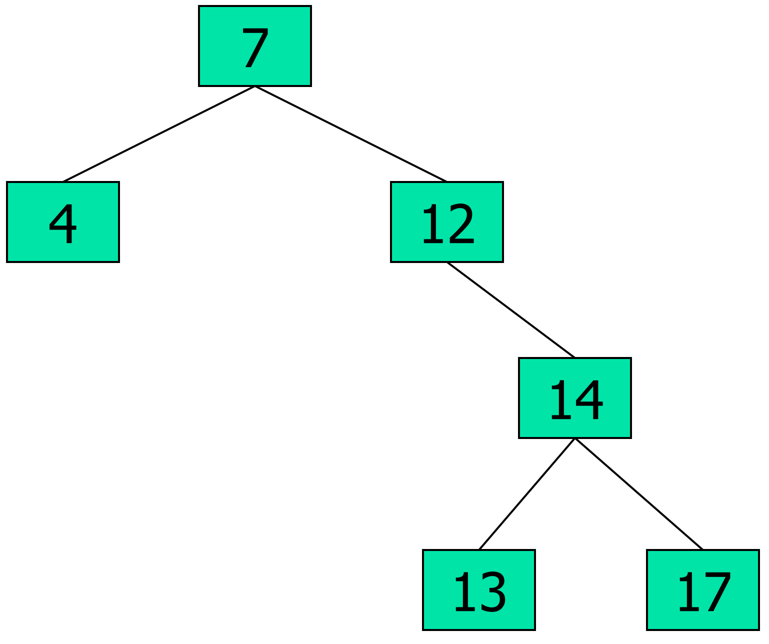


Eliminare din arbori AVL

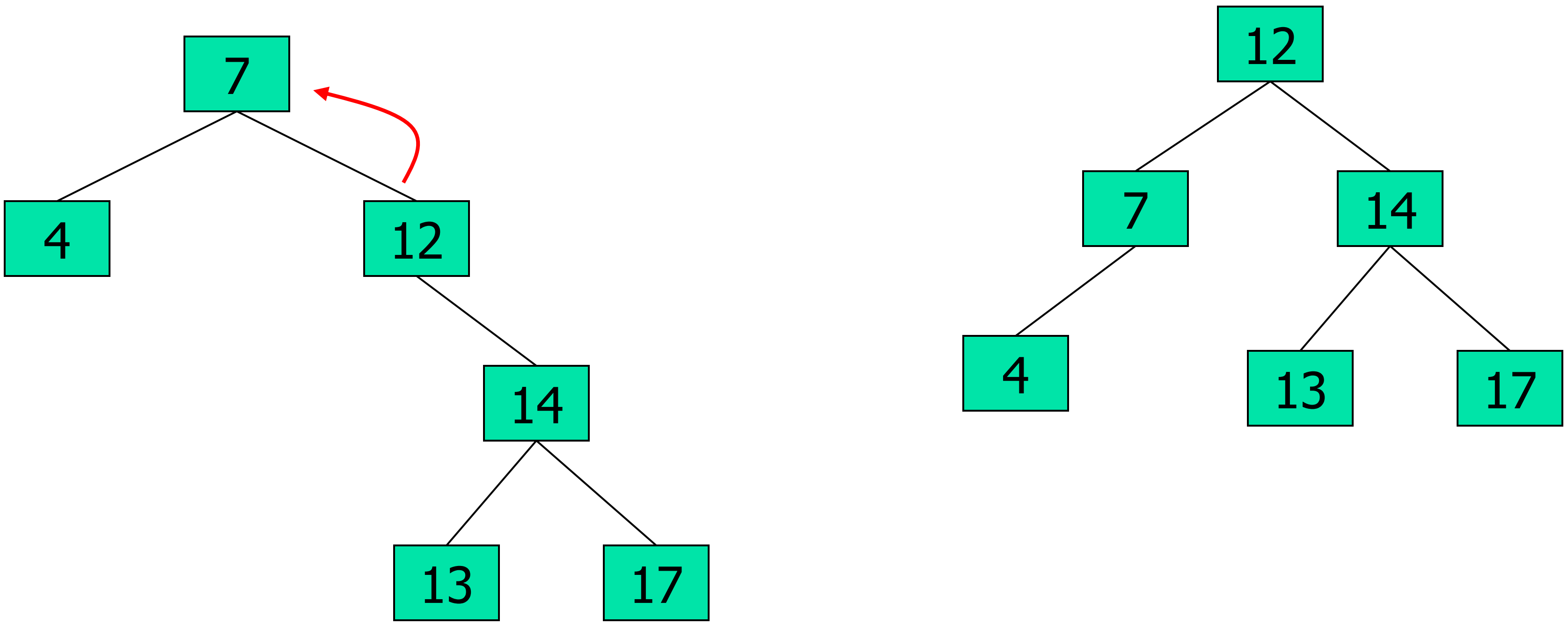


Eliminare din arbori AVL





Eliminare din arbori AVL



Inserare in arbori AVL

- Insereaza informatia ca o frunza (analog ca in arborii binari de cautare)
  - Nodurile - de la frunza adaugata la radacina arborelui - isi pot modifica factorul de echilibru
  - Se actualizeaza factorii de echilibru pe calea de la frunza inserata la radacina
  - Daca se obtin factori de echilibru 2 sau -2, atunci se reechilibreaza arborele folosind rotatii stanga / dreapta sau rotatii duble stanga / dreapta